Design Document: [App name]

Product: AI lost-and-found desk **Surfaces:** One mobile-first PWA with a bottom nav. After sign-in, the user's role shows either the claimant screens or the staff screens. **Direction:** Black and white minimalism, calm and precise, with Apple-like motion **Related:** User stories workbook · Architecture

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1. Principles

1. **One thing per screen.** Every form asks one question at a time, like Typeform. Every screen has one primary action.
2. **Content is the colour.** The UI is black, white and greys. The only colour comes from item photos, so the lost item is always the hero.
3. **Motion explains, never decorates.** Every animation answers "where did that come from?" or "did it work?". Nothing moves just to move.
4. **Calm under stress.** People open this app right after losing something. Copy is short, kind and specific. Never blame the user.
5. **Secrets stay secret.** The interface never hints at the private description. Not in placeholders, questions, errors or empty states.

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2. Design tokens

2.1 Colour

Pure greyscale. Semantic colour is used only for errors, and always next to text.

Token	Hex	Use
`--ink`	`#0A0A0A`	Primary text, primary buttons, active icons
`--ink-2`	`#3D3D3D`	Secondary text
`--ink-3`	`#737373`	Placeholder, captions, inactive icons (4.6:1 on white)

Token	Hex	Use
`--line`	`#E5E5E5`	Dividers, input borders
`--fill`	`#F5F5F5`	Input backgrounds, skeletons, cards on white
`--paper`	`#FFFFFF`	Page background
`--danger`	`#C62828`	Error text and borders only

Dark mode (should, not must): invert the scale with `--paper #0A0A0A` and `--ink #FAFAFA`, swapped under `prefers-color-scheme: dark`.

2.2 Typography

System fonts look native on every phone and cost zero bytes.

```
font-family: -apple-system, BlinkMacSystemFont, "SF Pro Text", "Inter",
"Segoe UI", Roboto, sans-serif;
```

Load **Inter** (400, 500, 600) from Google Fonts as the fallback on Android and desktop.

Token	Size / line height	Weight	Use
`display`	34 / 40	600	Question titles in step forms, success screens
`title`	22 / 28	600	Screen titles, item title
`body`	17 / 24	400	Body, inputs, chat
`callout`	15 / 20	500	Buttons, chips, list meta
`caption`	13 / 18	400	Helper text, timestamps, tab labels

All sizes are set in `rem` so browser text zoom works. Letter spacing is -0.4px on `display` and `title` only.

2.3 Spacing, radius, elevation

- **Spacing:** 4pt grid: 4, 8, 12, 16, 24, 32, 48. The screen gutter is 20px.
- **Radius:** 10 (inputs, chips), 14 (buttons), 20 (cards, sheets), 999 (pills, avatar).

- **Elevation:** no drop shadows on cards. Use a 1px `--line` border or a `--fill` background. Only floating layers (sheets, toasts, tooltips) get a soft shadow: `0 8px 32px rgba(0,0,0,0.08)`.
- **Touch targets:** at least 44 x 44px.

2.4 Iconography

Lucide icons at a 1.75px stroke and 24px size. The active tab uses the filled variant, or a 2.25px stroke when no filled version exists.

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3. Motion system

Motion is built with **Framer Motion** across the whole app (both roles). Only `transform` and `opacity` are animated, so everything runs on the GPU at 60fps.

3.1 Curves and timing

Token	Value	Use
`spring-snappy`	`{ type: "spring", stiffness: 500, damping: 35 }`	Button press, chips, toggles
`spring-smooth`	`{ type: "spring", stiffness: 260, damping: 30 }`	Page push and pop, sheets, cards
`ease-out`	`cubic-bezier(0.22, 1, 0.36, 1)`, 240ms	Fades, step transitions
`ease-in`	`cubic-bezier(0.4, 0, 1, 1)`, 180ms	Exits (always faster than entries)

3.2 Micro-interactions

Moment	Interaction
Any tap on a button or card	Scale to 0.97 on press, spring back on release (`spring-snappy`)

Moment	Interaction
Tab switch	Icon fills and scales 0.92 then 1. Content cross-fades in 180ms. No sliding between tabs.
Go deeper (list to detail)	New page slides in from the right 30% plus a fade. The previous page shifts left 10% and dims.
Item card to item detail	**Shared element:** the card photo grows into the detail carousel (`layoutId`).
Step form, next question	Current question slides up 24px and fades out. The next slides up from 24px below and fades in. The progress bar fills with `spring-smooth`.
Invalid submit	Field shakes horizontally (6px, 3 cycles, 300ms), the border turns `--danger` and the error text fades in. On Android, `navigator.vibrate(10)` adds a tick.
Search chips appear	Chips fade and rise 8px with a 40ms stagger. Removing a chip shrinks it to 0.8 and fades it out.
Chat message	The bubble scales from 0.96 and rises 8px. The bot shows a 3-dot typing indicator (dots pulse with 120ms offsets).
Claim submitted	Circle draws, then a checkmark stroke draws (400ms, SVG `stroke-dashoffset`), then the title fades in.
Bottom sheet	Slides up with `spring-smooth`; drag down to dismiss with rubber-band resistance; the backdrop fades to 40% black.
Status change in Requests	The chip text cross-fades and a dot badge pops (scale 0 to 1.15 to 1) on the tab icon.
Loading content	Skeletons in `--fill` with a slow shimmer (1.4s). No spinners for content; spinners appear only inside buttons.
Pull to refresh	A thin black arc fills as you pull, then rotates while loading.

3.3 Reduced motion

Under `prefers-reduced-motion: reduce`, all slides, scales and shared elements become 150ms cross-fades. Shimmer stops. The checkmark appears without drawing.

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4. Components

Component	Spec
Primary button	Full width on mobile, height 52px, radius 14, `--ink` background, white `callout` text. Disabled: `--line` background, `--ink-3` text. Loading: label replaced by a 16px white spinner, width unchanged.
Secondary button	Same size, white background, 1px `--ink` border.
Text button	`callout` in `--ink`, underlined on hover or focus.
Input	Height 52px, `--fill` background, no border at rest; on focus, a 1.5px `--ink` border and white background. Label sits above as `caption`. Helper text sits below in `--ink-3`; an error replaces it in `--danger`.
Big question input (step forms)	No box: 28px text with only a 2px bottom line, which turns `--ink` on focus.
OTP input	6 boxes of 48 x 56px with auto-advance; paste fills all of them. The focused box gets a 1.5px `--ink` border and a blinking caret.
Search bar	Height 48px, pill radius, `--fill` background, search icon on the left, clear (x) on the right when filled. Rotating placeholder cross-fades every 3s.
Chip	Height 32px, pill, 1px `--line` border, `callout`. Removable chips have an x. A dotted border means "not recognised" (with a tooltip).
Item card	1:1 photo with 20 radius, title (`callout` 600) and meta line (`caption` `--ink-3`: place and "2d ago"). Badge in the top-left corner: a white pill with black text ("Requested").
Status chip	Monochrome, distinguished by fill and icon, not colour: Submitted (outline, clock), Under review (outline, eye), Approved (solid black, check), Not selected or Rejected (`--fill`, minus), Collected (solid black, package).

Component	Spec
Bottom nav	Height 56px plus the safe-area inset, white with 80% opacity and `backdrop-filter: blur(20px)`, 1px top `--line`. Claimant: Home, Requests, Profile. Staff: Home, Items, Add (raised center button), Pickup, Profile.
Bottom sheet	20 radius top corners, 36 x 5px grab handle, max height 92dvh.
Toast	Black pill, white text, bottom center above the nav, auto-hides after 3s, swipe to dismiss.
Tooltip	(i) icon 16px next to the label. Tap or hover opens a black popover (max 260px wide, `caption` white text, 10 radius) with a 6px arrow. Closes on tap outside or Esc. Mobile tooltips never rely on hover.

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5. Validation and tooltip pattern

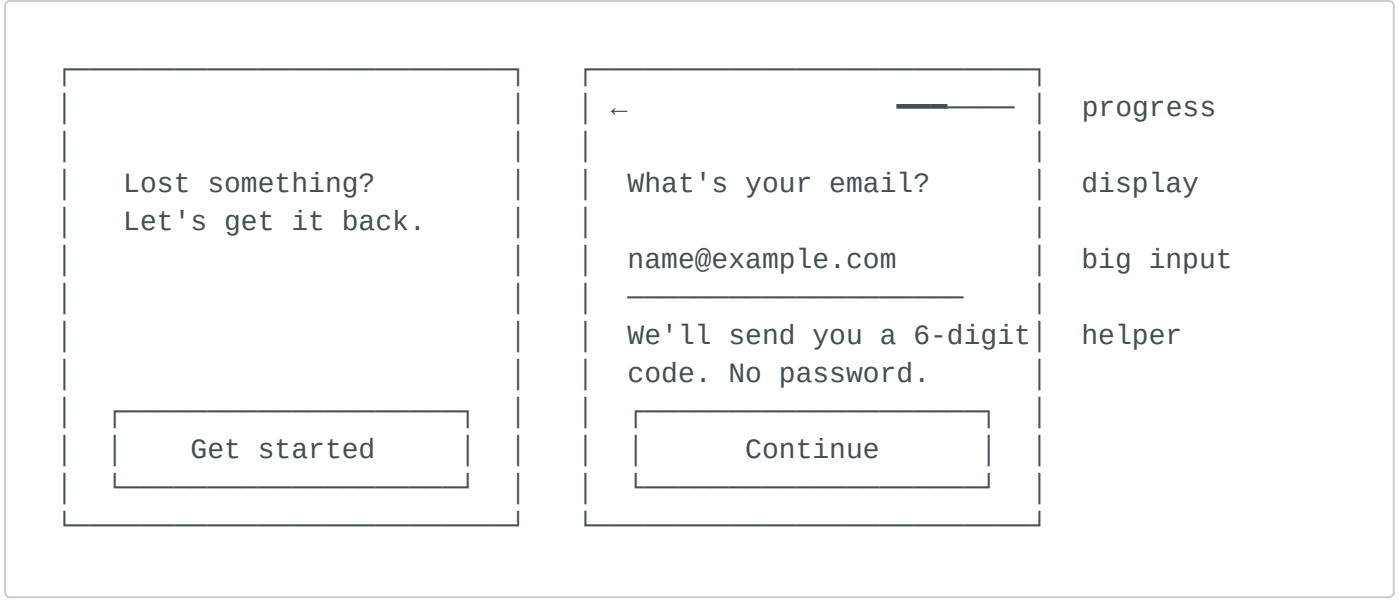
1. **Validate on blur, re-validate on change.** Errors never appear while someone is still typing the first time.
2. **The primary button stays disabled until the step is valid**, with a short hint under it explaining what's missing ("Add at least 2 photos").
3. **Errors are specific and kind.** Say what to do, not what went wrong: "Enter a valid email, like name@example.com," not "Invalid input."
4. **Tooltips explain why, helper text explains how.** Helper text is always visible under the field. A tooltip (i) is reserved for the "why" or privacy notes.
5. All error messages come from the **Validation Rules** sheet in the workbook, so web and PWA say the same thing.

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6. Claimant PWA screens

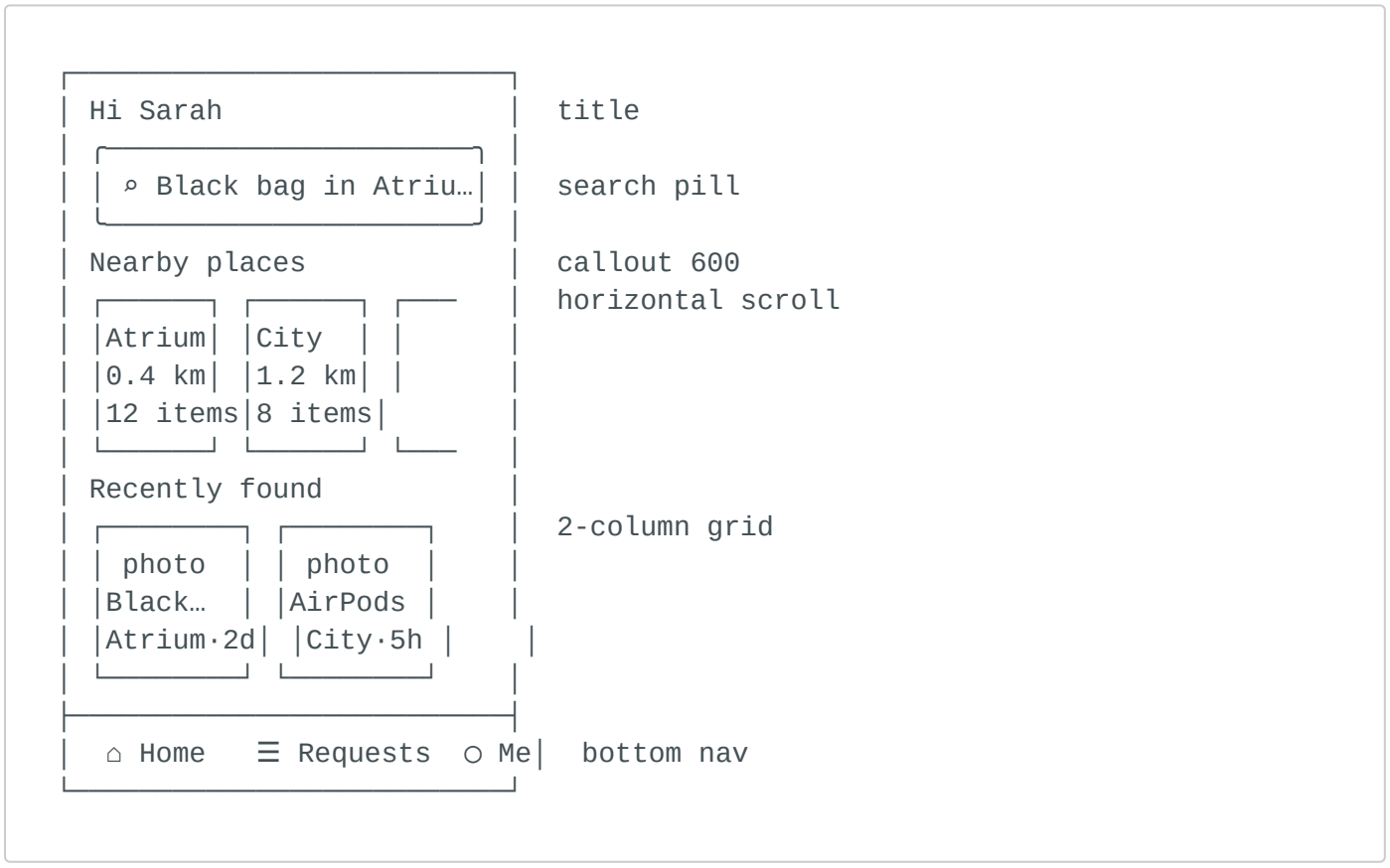
The layout is mobile-first, capped at 480px wide and centered on larger screens. Bottom nav: **Home · Requests · Profile**.

6.1 Onboarding (full screen, no nav)



Flow: Welcome, then Email, then Code, then Name (new users only), then Phone (new users only), then the Location primer, then Home. Each step uses the step-form transition. The keyboard opens automatically, and the Continue button sits just above it.

6.2 Home

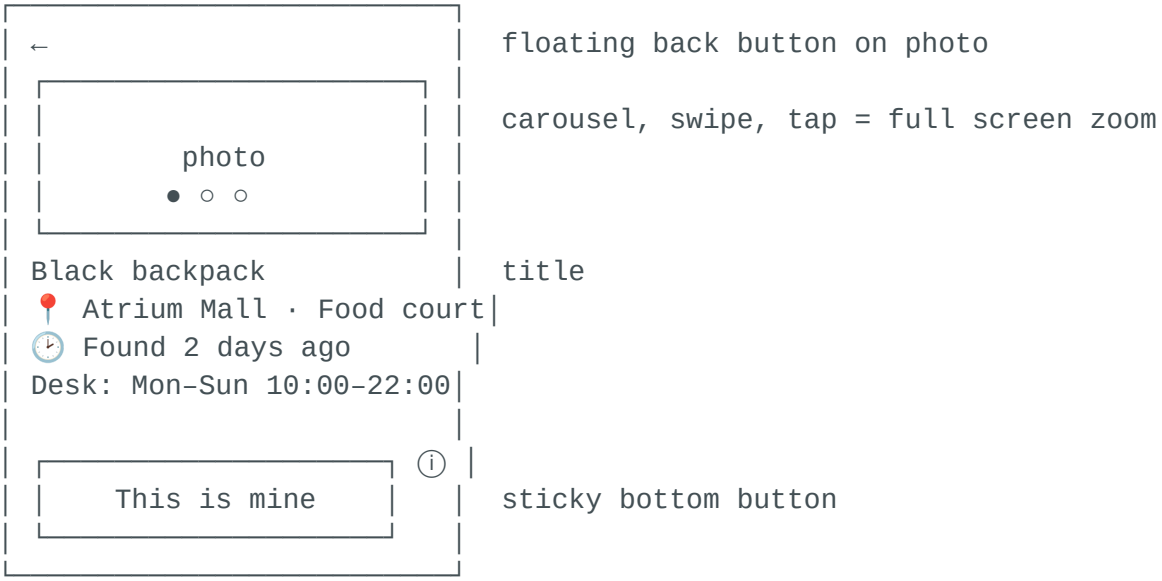


Tapping the search bar expands it to full screen: the bar moves to the top, the keyboard opens, and recent searches appear.

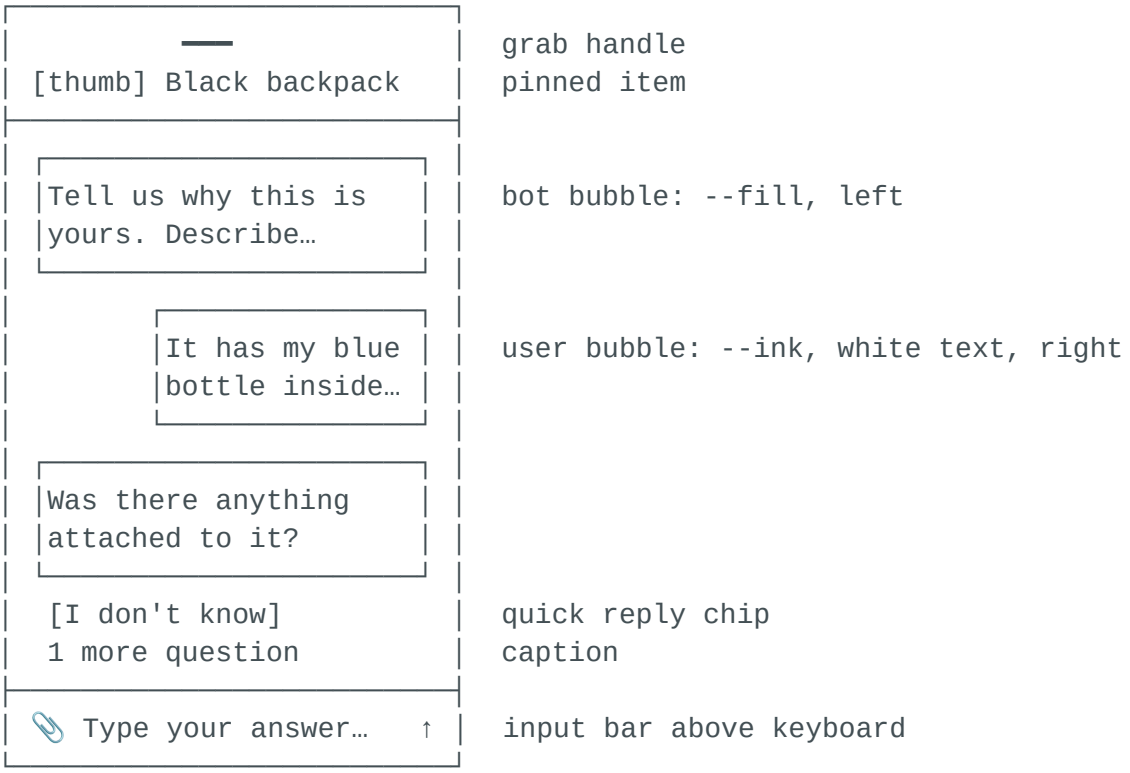
6.3 Search results

- The parsed chips sit under the bar: `[Black bag ×] [📍 Atrium Mall ×]`.
- A 2-column grid of item cards follows, with infinite scroll.
- Empty state: a centered line icon, "No match yet," and "Items are added all the time, so check back soon."

6.4 Item details



6.5 Claim chat (bottom sheet, 92% height)



The submitted state replaces the input bar with the success animation and a **See my requests** button.

6.6 Requests

- A list of rows: a 56px thumbnail, title, place, date, and a status chip aligned right.
- An approved row opens the **pickup card**: a large QR code on white, the 6-digit code (in tabular figures, grouped `482 913`), the place, desk hours, **Get directions** and "Valid until 14 Nov." Tapping the QR opens it full screen and turns on Wake Lock.

6.7 Profile

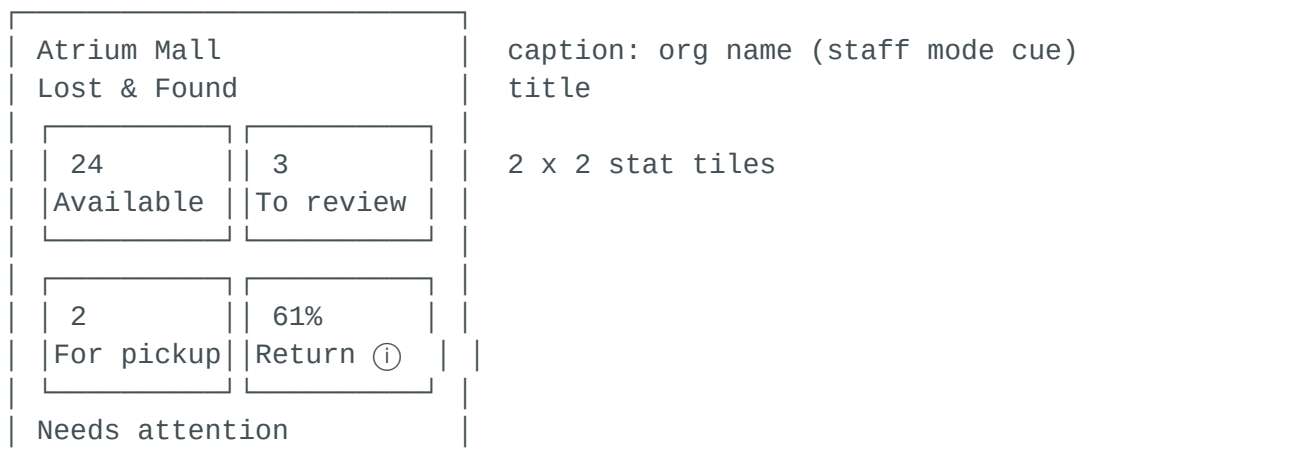
Grouped list (iOS settings style): name, email (read only) and phone, then the notifications and "Use my location" toggles, then Sign out and Delete account (text in `--danger`).

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7. Staff screens (same app, mobile-first)

Staff use **the same PWA**. They sign in with the same Email and Code screens, and their role sends them to the staff screens. The look, components and motion are identical; only the screens and the bottom nav change. On a laptop, staff see the same 480px phone layout, centered.

Staff bottom nav: Home · Items · **+ Add** (raised 56px black circle in the center) · Pickup · Profile



```
| [thumb] Black backpack 3 claims ›|
| [thumb] AirPods        1 claim  ›|
├──────────────────────────────────┤
| ⌂ Home  ▦ Items  ( + )  # Pickup  ○ |
└──────────────────────────────────┘
```

Screen	Layout
Staff Home	The org name as a caption above the title (so staff always know they're in staff mode). 2 x 2 stat tiles (big number in `display`, label in `caption`), with the return rate in the fourth tile. Below: a "Needs attention" list of items with claims waiting for review.
Add (center tab)	Opens the full-screen step form, sliding up like a sheet. Steps: Photos (a big "Take photo" tile that opens the rear camera directly, a "Choose from gallery" link, and a thumbnail strip), Title (AI suggestion label), Location, Private description (lock icon), Review. The success screen offers "Add another" (back to the camera) and "View item".
Items	Horizontally scrolling status pills, a search bar, then a 2-column card grid. Each card has a status chip and a claim-count badge. The location filter opens in a bottom sheet.
Item page	Photo carousel, details, then segmented tabs: **Claims · Details · Activity**. Claims are sorted by score; each row has a score bar (a black fill on a `--fill` track, with the number and a High, Medium or Low label). Tapping a claim opens a bottom sheet with the AI rationale checklist (✓ matched, ½ partial, ✕ missed, ⚠ contradiction, each written out in words), the chat transcript, and sticky **Approve** and **Reject** buttons.
Pickup tab	Opens straight into a full-screen camera QR scanner with a rounded square guide and a dimmed surround, plus an "Enter code instead" button that swaps in the 6-digit input. A valid scan triggers a short vibration (Android) and slides up a confirmation sheet with the item photo, claimant name, phone ending, and **Hand over**.
Profile	Name, organization, Sign out.

8. Voice and microcopy

- Sentence case everywhere. No exclamation marks except "Good news!" in the approval notification.
- Use "your item" and "we", never "the user" or "the system".
- Keep it under 12 words where possible.
- **Never** confirm or deny a private detail. Not "Correct!" or "That's not right," only "Thanks, got it."

Situation	Copy
Claim submitted	"Thanks, your request was submitted. We'll let you know."
Not selected	"This item went to another owner. We hope you find yours soon."
Rejected	"We couldn't verify this item as yours. If you think this is a mistake, visit the desk."
Approved push	"Good news! Your black backpack is ready for pickup at Atrium Mall."
Offline	"You're offline. Showing saved results."

9. Accessibility

- Text contrast is at least 4.5:1. The greyscale palette above is checked against white.
- Status is never shown by fill alone. Every status chip has an icon and a word.
- Focus rings: a 2px `--ink` outline with a 2px offset on every interactive element (`:focus-visible`).
- The bottom nav is a `<nav>` with `aria-current="page"` on the active tab. Sheets trap focus and close on Esc.
- Chat uses `aria-live="polite"` so screen readers announce new bot messages.
- Respect `prefers-reduced-motion` (section 3.3) and browser zoom up to 200%.